DERIVED ADJOINT PAIR FROM THE LQF LOWER SHRIEK AND UPPER SHRIEK

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Definition of f_! for locally quasi-finite morphisms (Definition definition-f-shriek-lqf, canonical tag 0F6R). cited at statement line 2588. The left adjoint being derived.
- Upper shriek for locally quasi-finite morphisms (Lemma lemma-lqf-f-upper-shriek, canonical tag 0F59). cited at statement lines 2588-2589. The right adjoint being derived.
- Derived adjoint functors (Derived Categories, Lemma derived-lemma-derived-adjoint-functors, canonical tag 09T5). cited at proof lines 2600-2601. Produces the adjoint pair on derived categories from the adjoint pair on abelian categories.
- Stalks of f_! for locally quasi-finite morphisms (Lemma lemma-lqf-f-shriek-stalk, canonical tag 0F5F). cited at proof lines 2602-2603. Gives exactness of f_!, hence Lf_! = f_!.

Source reference. Original definition 0F6R: sources/stacks-chow/tags.txt, source lines 18981–18981. Cited at statement line 2588: the left adjoint being derived.

Source reference. Original lemma 0F59: sources/stacks-chow/tags.txt, source lines 18929–18929. Cited at statement lines 2588-2589: the right adjoint being derived.

Source reference. Original lemma 09T5: sources/stacks-chow/tags.txt, source lines 12378–12378. Cited at proof lines 2600-2601: produces the adjoint pair on derived categories from the adjoint pair on abelian categories.

Source reference. Original lemma 0F5F: sources/stacks-chow/tags.txt, source lines 18935–18935. Cited at proof lines 2602-2603: gives exactness of f_!, hence Lf_! = f_!.

Lemma 0.2. *Let $f : X \to Y$ be a locally quasi-finite morphism of schemes. Let Λ be a ring. The functors $f_!$ and $f^!$ of Definition 0F6R and Lemma 0F59 induce adjoint functors $f_! : D(X_{étale}, \Lambda) \to D(Y_{étale}, \Lambda)$ and $Rf^! : D(Y_{étale}, \Lambda) \to D(X_{étale}, \Lambda)$ on derived categories.*

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Proof. This follows immediately from Derived Categories, Lemma 09T5, the fact that $f_!$ is exact (Lemma 0F5F) and hence $Lf_! = f_!$ and the fact that we have enough K-injective complexes of Λ-modules on $Y_{étale}$ so that $Rf^!$ is defined. □